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Histogram of an images

Aim

To obtain a histogram for finding the frequency of pixels in an Image with pixel values ranging from 0 to 255. Also write the code using OpenCV to perform histogram equalization.

Software Required:

Algorithm:

Step 1:

Read the gray and color image using imread()

Step 2:

Print the image using imshow().

Step 3:

Use calHist() function to mark the image in graph frequency for gray and color image.

step 4:

Use calHist() function to mark the image in graph frequency for gray and color image.

Step 5:

The Histogram of gray scale image and color image is shown.

Program :

Developed By : VISVESWARRAN H

Register Number : 212224110063

Input Grayscale Image :

```
import cv2
from matplotlib import pyplot as plt
# Load the color image
image = cv2.imread('image.jpg')
# Convert the image to grayscale
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
plt.imshow(gray_image, cmap='gray')
plt.title('Original Grayscale Image')
plt.axis('off')
```



Histogram of Grayscale Image :

```
hist_original = cv2.calcHist([gray_image], [0], None, [256], [0, 256])  
plt.plot(hist_original, color='black')  
plt.title('Original Histogram')  
plt.xlim([0, 256])
```



Equalization :

```
# Apply histogram equalization  
equalized_image = cv2.equalizeHist(gray_image)  
plt.imshow(equalized_image, cmap='grey')  
plt.title('Equalized Image')  
plt.axis('off')
```



Equalized Histogram :

```
hist_original = cv2.calcHist([equalized_image], [0], None, [256], [0, 256])  
plt.plot(hist_original, color='black')  
plt.title('Equalized Histogram')  
plt.xlim([0, 256])
```



Output:

Input Grayscale Image and Color Image

```
[12]: (np.float64(-0.5), np.float64(2047.5), np.float64(1364.5), np.float64(-0.5))
```

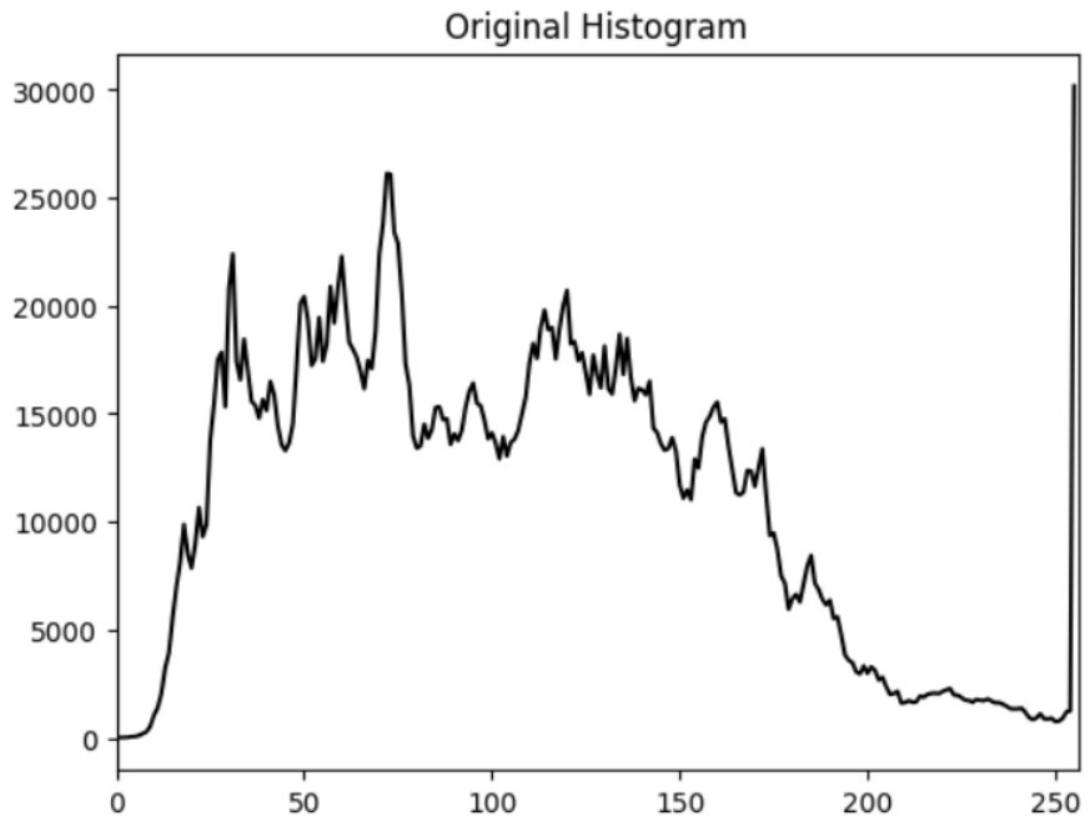
Original Grayscale Image





Histogram of Grayscale Image and any channel of Color Image

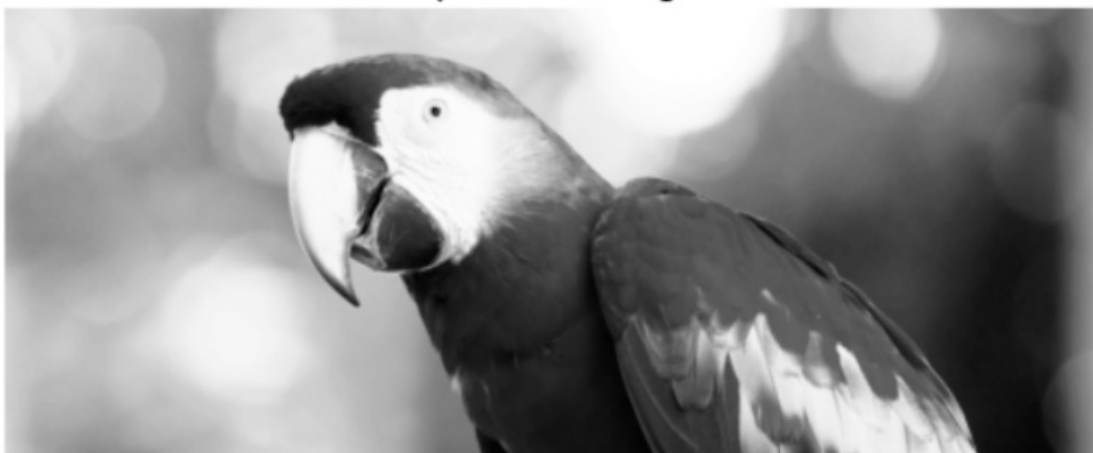
```
[13]: (0.0, 256.0)
```



Histogram Equalization of Grayscale Image.

```
(np.float64(-0.5), np.float64(2047.5), np.float64(1364.5), np.float64(-0.5))
```

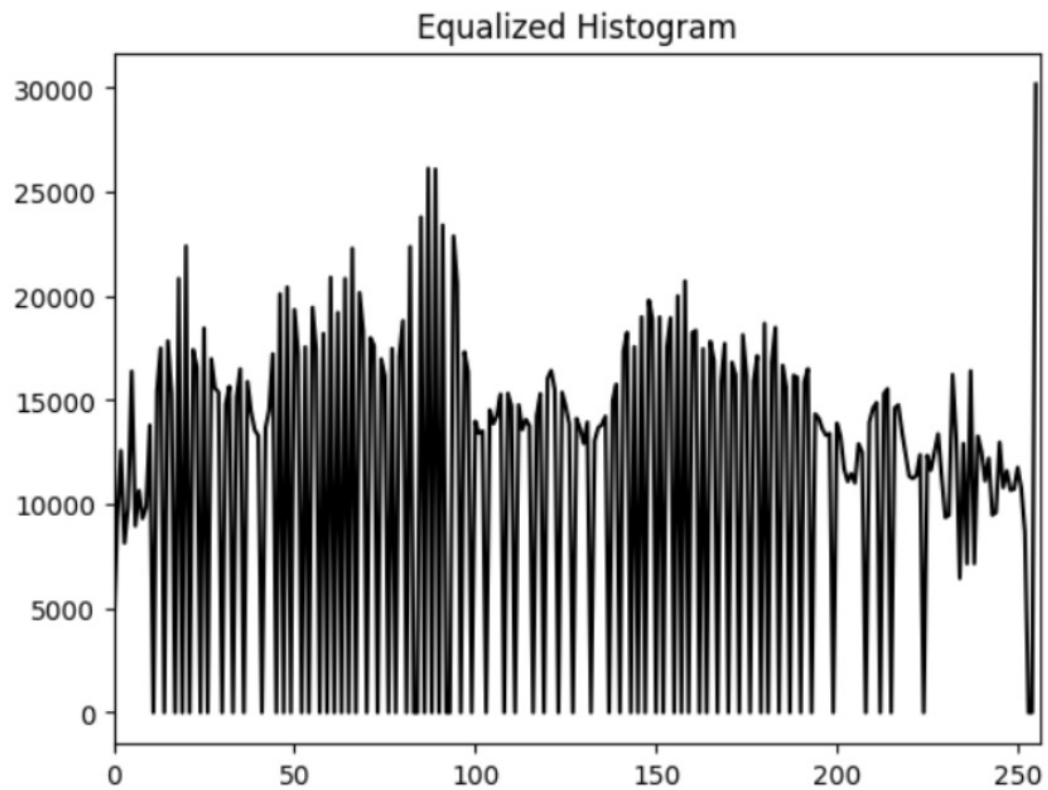
Equalized Image





Equalized Histogram.

[22]: (0.0, 256.0)



Result:

Thus the histogram for finding the frequency of pixels in an image with pixel values ranging from 0 to 255 is obtained. Also, histogram equalization is done for the gray scale image using OpenCV.

No releases published

Packages

No packages published

Languages



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